

BAAZ & UDAY JKBOSE

2026 CLASS 11TH



ANIMAL KINGDOM

LECTURE-3

ZOOLOGY

DAVOOD SIR (CEO of NG)

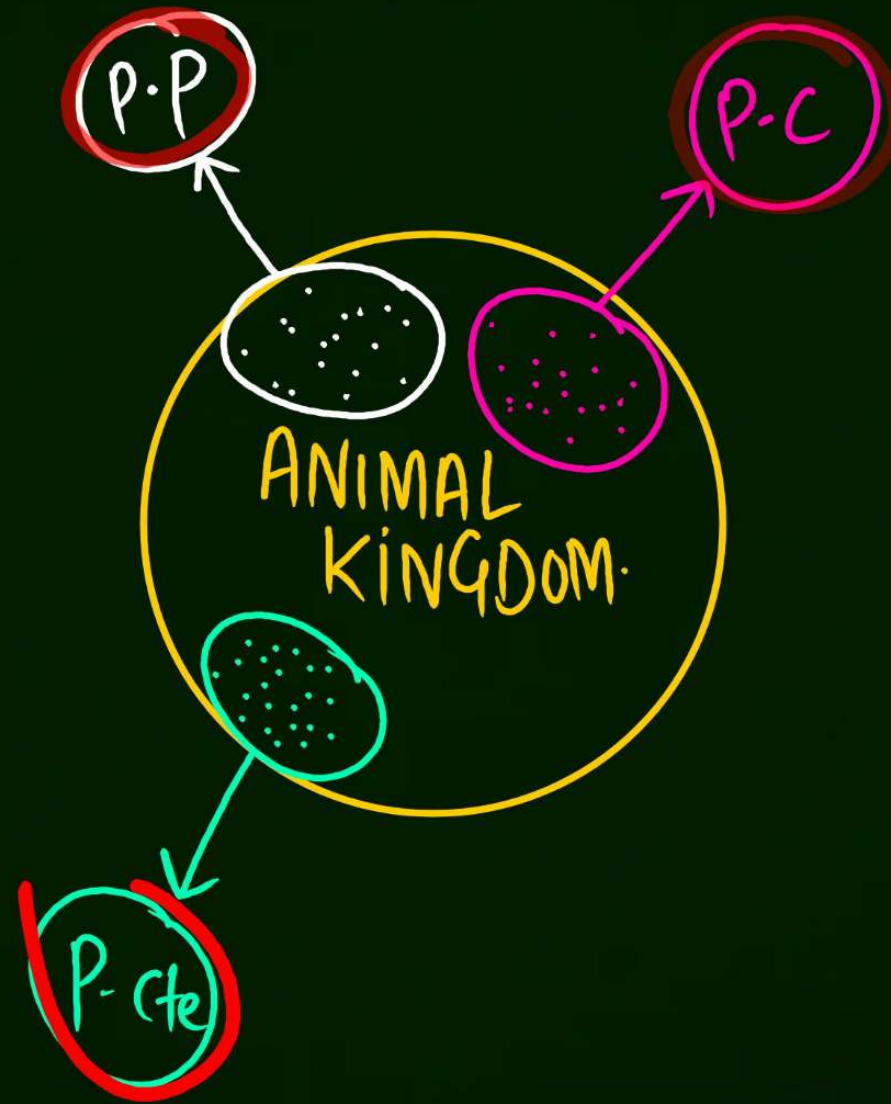


Physics Wallah

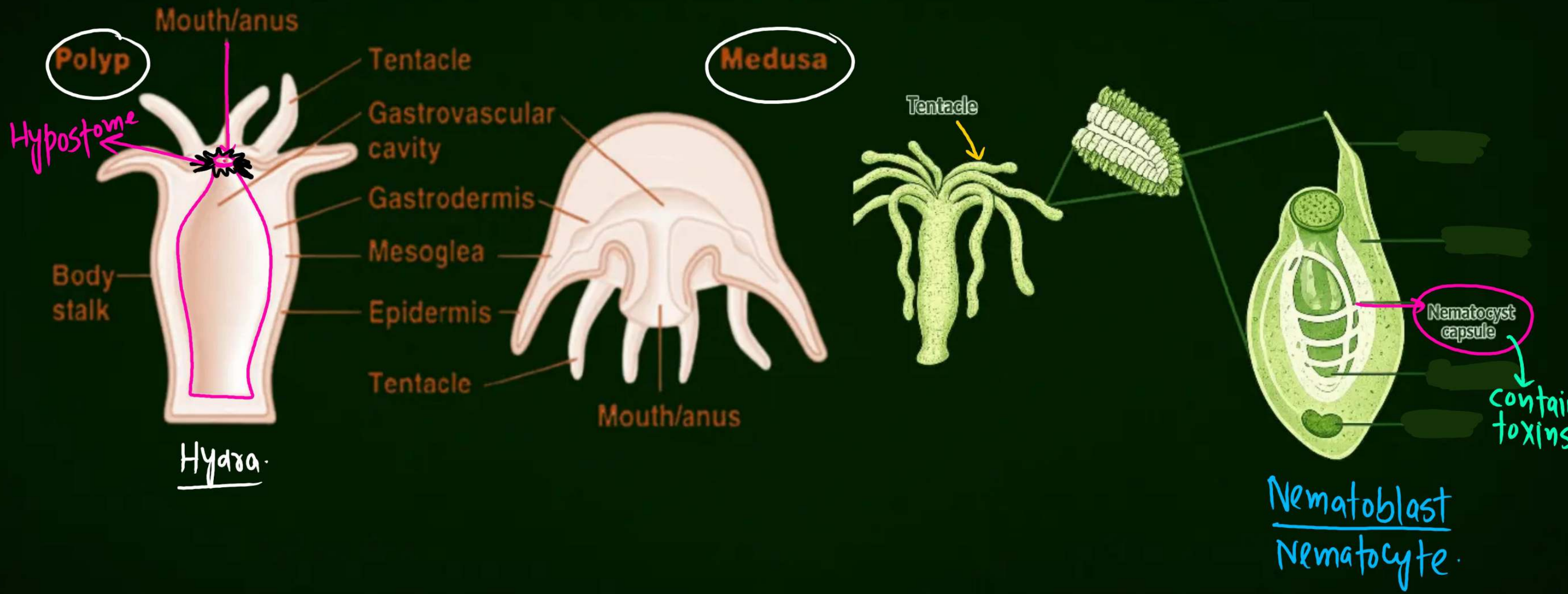
Topics To Be Covered

- 1 **Phylum Coelentrata**
- 2 **Phylum Ctenophora**
- 3 **Previous Year NEET MCQ's**

Kingdom Animalia



Phylum Coelenterata (Cnidaria)



Phylum Coelenterata or Cnidaria (Sea Stingers)

Habitat:- Aquatic (mostly marine), Some are Sessile while many are free swimming. they have the ability of stinging.

Symmetry & Body Organisation: Radially symmetrical, Diploblastic, Tissue level of organization.

Digestion:- Both intracellular and extracellular digestion

Skeleton:-

Gastro-vascular Cavity:- Posses a Central cavity called gastrovascular cavity or Coelenteron-having a hole acts as both mouth & anus located above the hypostome.

Special Feature:- → The name Cnidaria comes from Cnidoblast/Cnidocyte (these are the special cells containing nematocyst (stinging capsule) present in tentacles of body.

Functions of Nematocysts: for Defense, anchorage and prey capturing.

Phylum Coelenterata or Cnidaria (Sea Stingers)

Cnidarians have two main body forms:

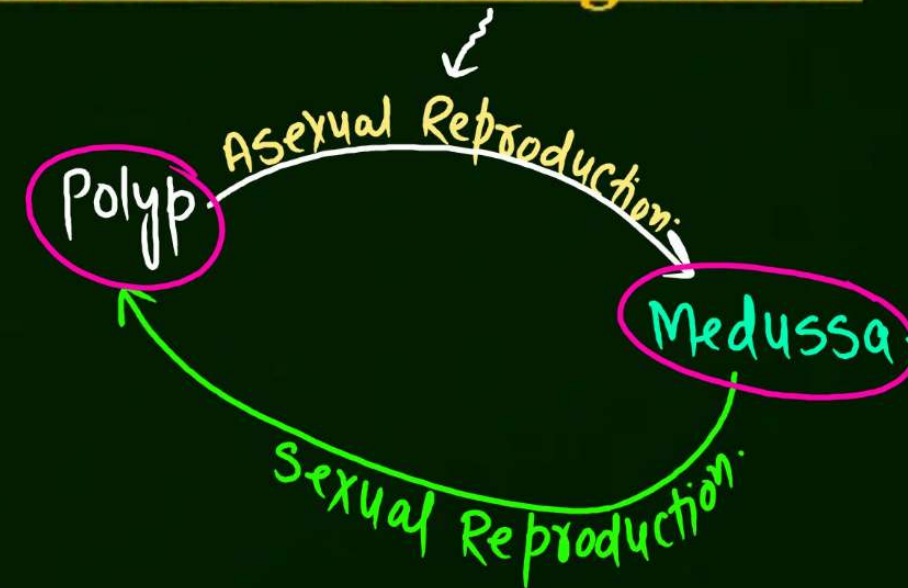
- Polyp:- they are cylindrical and sessile e.g. Hydra and Adamsia
- Medusa:- they are umbrella shaped but free swimming e.g. Aurelia & Jellyfish

Some cnidarians show alternation of generations called Metagenesis:

- Polyps form Medusa by Asexual Reproduction.
- Medusa form Polyps by Sexual Reproduction.

Examples:

Physalia	–	Portuguese man-of-war
Adamsia	–	Sea anemone
Pennatula	–	Sea pen
Gorgonia	–	Sea fan
Meandrina	–	Brain coral



Phylum Coelenterata (Cnidaria) Mnemonic

DC'S R GEM

- **D** → Diploblastic
- ✓ • **C** → Cnidoblasts
- **R** → Radial Symmetry
- **G** → Gastrovascular cavity
- **E** → Examples: Hydra, Aurelia
- ✓ • **M** → Metagenesis

Phylum Ctenophora (Luminous Marine Gliders)



Phylum Ctenophora (Luminous Marine Gliders)

Habitat:- they are also called as Sea Walnuts or Comb jellies.
they are exclusively marine and show elegant movement with glowing body.

Symmetry & Body Organisation:- Radially symmetrical and diploblastic having tissue level of organisation.

Digestion:- Both intracellular and extracellular

Unique Characters (Nabkaar Character):-

- Body surface bears distinct rows called ciliated Comb plates (8 in number) helps in locomotion.
- they exhibit bioluminescence means they emit light naturally.

Reproduction & Development:-

- they are bisexual (Hermaphrodites)
- External fertilization and indirect development (Larval stage).

Examples: *Pleurobrachia* – the classic sea gooseberry.

Ctenoplana - Comb Jelly or Sea Walnut

Phylum Ctenophora (Mnemonic)

"CBR" – Comb, Bioluminescence, Radial

C – Comb plates (8 rows of cilia for movement)

B – Bioluminescent (they glow in the dark)

R – Radial symmetry

Thank
You